Smart AI powered Agriculture System

Muhammad Awais (BSE223112)
Hamza Bashir (BSE223108)
Junaid Amin (BSE223107)

Supervised by: Adnan Karamat

Fall 2025

Department of Software Engineering

Capital University of Science & Technology, Islamabad

Table of Contents

1. Introduction 2

1.1. Project Introduction 2

1.2. Existing Examples / Solutions 3

1.3. Problem Statement 3

1.4. Business Scope 4

1.5. Useful Tools and Technologies 6

1.6. Project Work Break Down 7

1.7. Project Time Line 8

List of Tables

Table 1-1 Existing Examples 3

Table 1-2 Lean Canvas 5

List of Figures

Figure 1-1 Work Breakdown 7
Figure 1-2 Gantt Chart 8

1. Introduction

Agriculture in Pakistan faces challenges like water shortage, low productivity, and lack of modern tools. To address this, our project **Smart AI-Powered Agriculture System** uses IoT sensors to monitor soil and weather conditions, automates irrigation to save water, and applies AI to recommend suitable crops. This solution helps farmers improve yield, reduce costs, save water and adopt smarter farming practice.

1.1. Project Introduction

Our project, **Smart AI-Powered Agriculture System**, is an IoT-based solution designed to make farming more efficient and sustainable. The system uses sensors to monitor soil moisture, temperature, humidity, light, and rainfall in real time. This data is sent to a mobile app through a backend server, allowing farmers to track field conditions, receive alerts, and view historical records. The project also includes **automatic irrigation**, where water is supplied only when the soil is dry, and an **AI module** that recommends the most suitable crops for upcoming seasons. The goal is to help farmers conserve water, improve crop yield, and make data-driven decisions in agriculture.

The main **beneficiaries** of this project are **small and medium scale farmers in Pakistan**, who often face challenges like water wastage, low crop productivity, and lack of access to modern agricultural tools. By using our system, these farmers can monitor their fields easily, save water through automated irrigation, and receive crop recommendations that support better planning. This helps them reduce costs, increase income, and adopt smarter farming practices, ultimately improving their livelihood.

1.2. Existing Examples / Solutions

Table 1-1 Existing Examples

Name	What they do (features)	What gap remains
SAWiE (Sustainable Agriculture Water & Intelligent Ecosystem)	Offers IoT + ML-based advisory for farmers. Soil mapping & alerts for drought/floods. (SAWiE Ecosystem)	They don't automate irrigation hardware/pump control directly
Buraq Integrated Solutions — Smart Drip Irrigation	They implement drip irrigation systems with soil moisture & temperature sensors. (Buraq Integrated Solutions (Pvt) Ltd)	Lacking AI Crop recommendation module
VGreen — CropSight™	Farm mapping, real-time monitoring, analytics. Localized for Pakistan's farms. (vgreen.com.pk)	Good dashboards and analytics, but lacks full automation (pump control, crop recommendation) and is costly.
Field Commander — Valley Irrigation Pakistan	Remote monitoring and SMS/Email alerts (valleyirrigationpakistan.com)	They don't offer smart irrigation and only available for large farms.
Smart IoT Farm at PMAS-AAUR, Rawalpindi	University research farm with sensors (soil moisture etc.), real-time monitoring via IoT, to demonstrate precision agriculture to farmers. (The News International)	There is no proper mobile app for farmers, no AI decision support, and no automation for irrigation.

1.3. Problem Statement

Agriculture is the main source of income and food in many countries, especially in developing regions. Most farmers depend on farming for their livelihood, but they face many challenges such as wasting water, unpredictable weather, no access to real-time information, and no proper guidance for choosing crops or managing their fields. Traditional farming methods often cause problems like over-watering or under-watering, loss of soil nutrients, and low crop production.

Even though modern technology is available, many small and medium farmers cannot afford or access smart farming tools. There is a need for an affordable and easy-to-use smart agriculture system that uses IoT and AI to monitor the environment, save resources like water and energy, and give farmers useful recommendations. Such a system can help farmers make better decisions, increase crop yield, and practice farming in a more sustainable way.

1.4. Business Scope

Our project, **Smart AI-Powered Agriculture System**, has good potential in the agriculture sector of Pakistan. Since most farmers face problems like wasting water, low crop yield, and lack of modern tools, this system provides them with an **affordable and simple solution**. By selling **IoT-based smart farming kits** (sensors + controller + app), farmers can save water, get better crop production, and make smarter decisions.

This system can be used by **small and medium farmers**, and it can also be promoted through **government programs, NGOs, and agriculture companies**. The idea has strong business value because it directly improves income of farmers and supports food security.

Table 1-2 Lean Canvas

Problem	**Solution**	**Unique Value Proposition**	**Unfair Advantage**
1. Water wastage due to over-irrigation. 2. Low crop yield from poor crop choice. 3. Lack of affordable smart farming tool.	1. IoT sensors for soil and weather. 2. Automatic irrigation system. 3. Mobile app with AI-based crop suggestion	An affordable smart farming kit that saves water, increases yield, and helps farmers make better crop decisions.	1. Localized solution for Pakistani crops. 2. Low cost compared to imports. 3. Easy to use for non-technical farmers.
Existing Alternatives	**Key Metrics (Approx.)**	**High-Level Concept**	**Channels**
1. Manual irrigation. 2. Expensive imported smart farming systems.	1. 100+ farmers will use the system. 2. ~ 45% of water saved. 3. ~ 35% increase in crop yield.	A smart farming assistant for every farmer, like a personal digital guide for crops.	1. Direct sales to farmers. 2. Partnerships with NGOs & govt programs. 3. Agriculture supply companies.
Cost Structure	**Customer Segments**	**Early Adopters**	**Revenue Structure**
1. IoT hardware (sensors, controllers). 2. Mobile app & server hosting. 3. Installation & support.	1. Small & medium farmers. 2. Agriculture NGOs & government projects.	1. Farmers with 1–4 acres land. 2. NGOs running pilot agriculture projects.	1. Selling hardware kits. 2. Subscription for app features. 3. Installation/training services.

1.5. Useful Tools and Technologies

Hardware

- **ESP32** – Microcontroller with Wi-Fi for sensor data processing & transmission.
- **Soil Moisture Sensor (YL-69 / Capacitive Sensor)** – Detects soil water level.
- **DHT22 Sensor** – Measures temperature & humidity.
- **Light Sensor (LDR)** – Monitors sunlight levels for crop growth.
- **Rain Sensor (FC-37 / YL-82)** – Detects rainfall to avoid unnecessary irrigation.
- **Relay Module** – Controls water pump/valve automation.
- **Water Flow Sensor** – Measures water usage during irrigation.

Software

- **Arduino IDE using C++** – For programming the ESP32
- **Backend Server:** NodeJS – Handles data from sensors.
- **Database:** Firebase Realtime Database / MySQL – Stores sensor readings & history.
- **Mobile App:** Android (Java/Kotlin) OR Flutter (cross-platform) – Displays live data, history, and alerts.
- **Data Visualization:** Graphs & charts (MPAndroidChart / Chart.js / Recharts).
- **Notifications:** Firebase Cloud Messaging (push alerts)

AI & ML

- **Python (TensorFlow / Scikit-learn)** – For crop recommendation & predictive analytics.
- **Signal Processing** – To clean noisy sensor data before analysis.

1.6. Project Work Break Down

The project breakdown divides the overall work into smaller, manageable components to ensure systematic execution and clarity of responsibilities. Each major task is broken down into sub-tasks, and further into smaller actionable sections, defining specific deliverables and timelines. This structured approach helps in effective planning, resource allocation, progress tracking, and achieving project objectives efficiently. Our project is divided into several parts as shown in Figure 1-1.

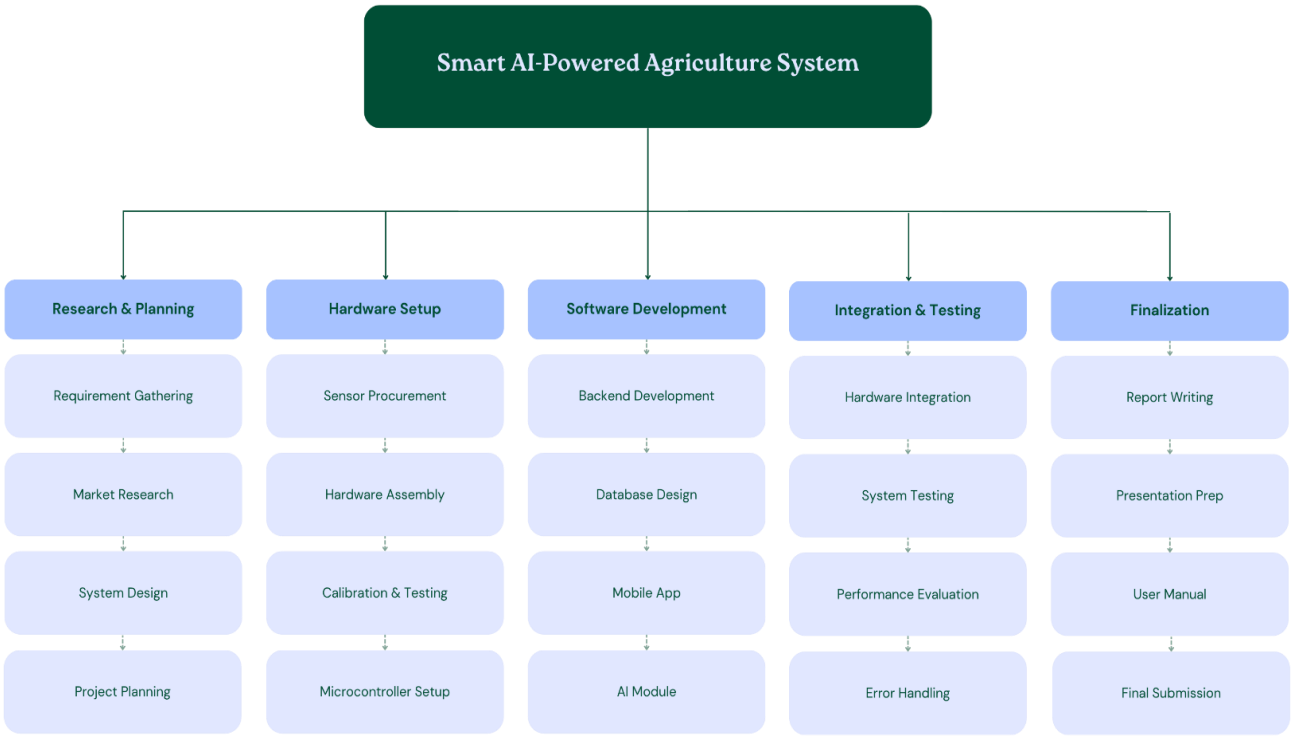

Figure 1-1 work breakdown

1.7. Project Time Line

The project timeline outlines the planned phases from **29 September 2025 to 10 July 2026**, covering research, hardware setup, software development, and final integration. Each phase is scheduled based on workload to ensure smooth progress — from requirement gathering and system design to implementation, testing, and final presentation. A Gantt chart is shown in Figure 1-2.

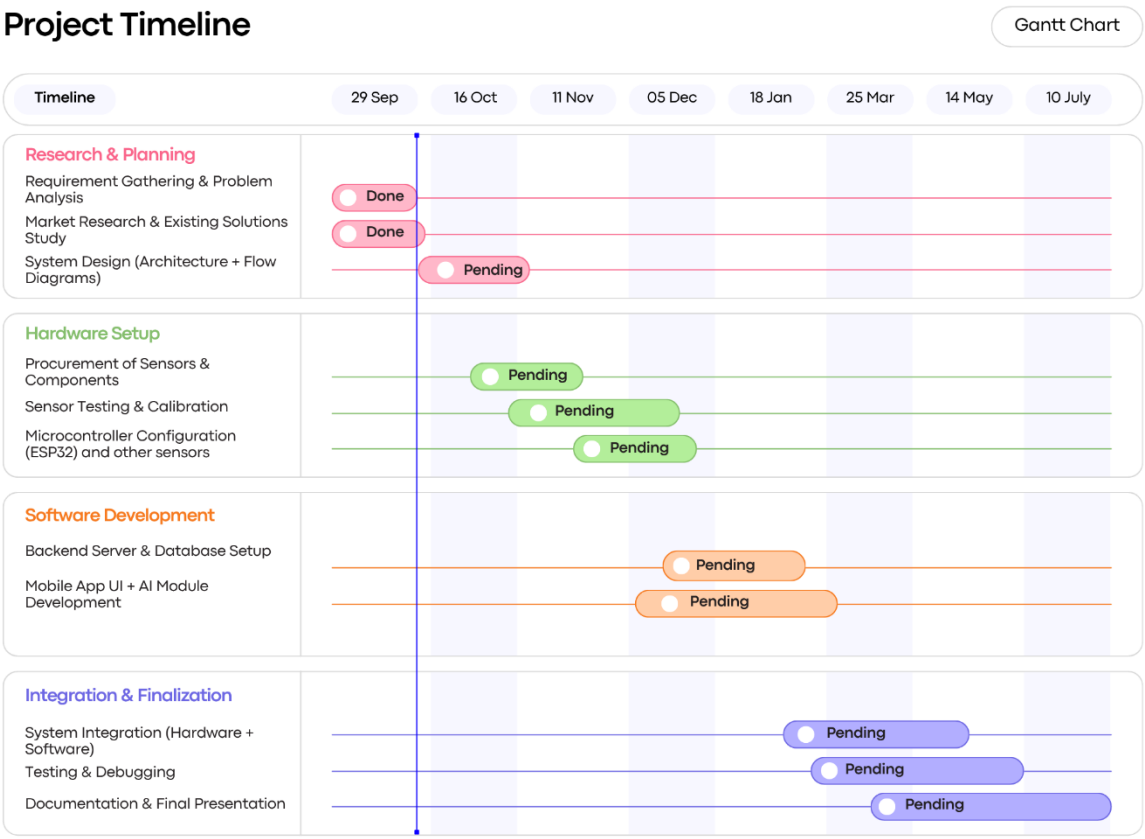

Figure 1-2 Gantt chart